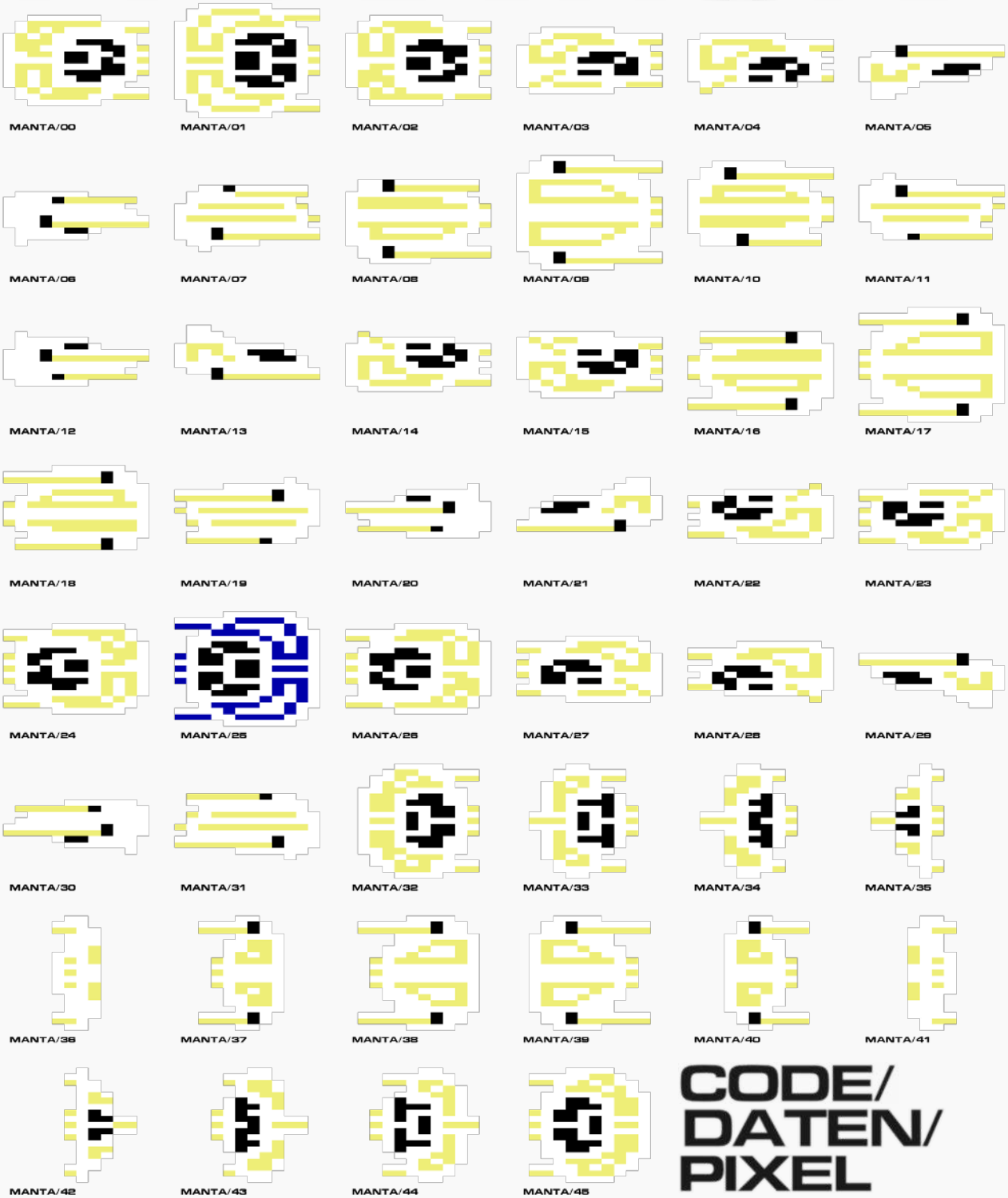
URIDIUM/ FRAKTION



CODE/ DATEN/ PIXEL

URIDIUM/ FRAKTION

For Edna.

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ABOUT/THIS/BOOK

This book isn't about the history of *Tempest* or its importance in the history of computer games. This is a book about computer code. Each chapter aims to be an easily digested sketch of how a specific feature or effect was achieved. In the case of *Tempest* this means sifting through the 6502 assembly code David Theurer wrote by hand(!) to create the game, and wrangling with the bits and bytes of Atari's Quadrascan Vector Graphics. For *Tempest 2000* it means familiarizing ourselves with Motorola 68K assembly code and the Jaguar's powerful hardware platform.

There are lots of fascinating bits and pieces to pore over. And it is interesting to see how the years that separate the two games leave some things unchanged and many other things changed utterly. I've put together this book of squibs in the hope they will allow you, the reader, to digest small bite-sized insights into each game. And I hope, by interleaving elements from *Tempest* and *Tempest 2000*, to enable you to compare the parallel lives in the design and execution of separate incarnations of the same classic video game.

Rob Hogan (@mwenge)
2026

URIDIUM/
FRAKTION

MANTA/SPIN

When waiting to start the first level, the player's Manta ship spins elegantly in the center of the screen. This simple animation uses an enormous number of sprites - 16 in total.

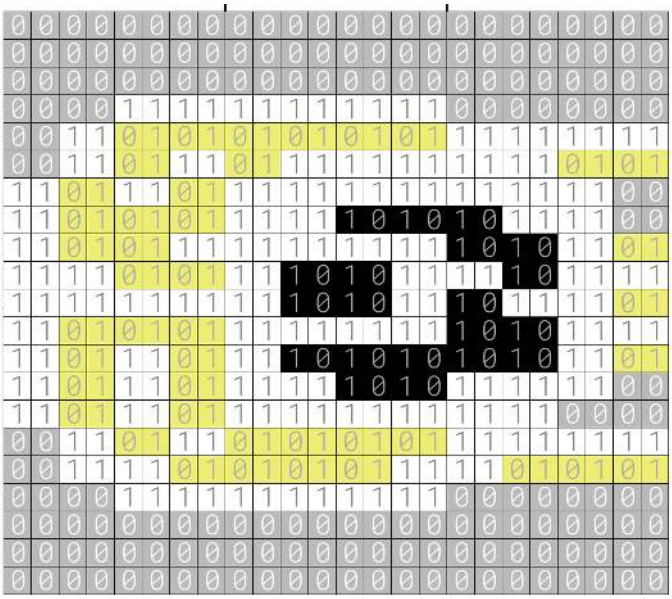
The first phase of the spin - - does not start from where we might expect. The rotation of the initial ship is negative so that we start from a slightly rotated position.

The next frame in the animation - - rotates the ship to the opposite roll position.

So the Manta starts tilted away from us, rotates to a flat position, and then begins its roll towards us. Notice the subtly different assymetry in the cockpit of the Manta when it is rolled away from us compared to when it is rolled towards us. It is easiest to compare this difference in detail on the following page where we blow up each of the tilted sprites to see how they are constructed from their source data.

URIDIUM/MANTA/00

SPEC:MANTA
AZIMUTH:090°
OFFSET:$40
ELEVATION:000°
RAM:$5000
ROLL:015°

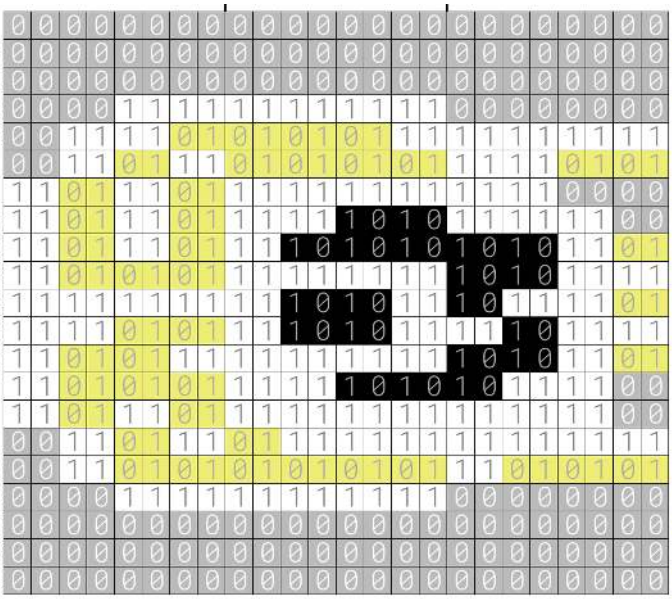


```
.BYTE $00,$00,$00
.BYTE $00,$00,$00
.BYTE $00,$00,$00
.BYTE $0F,$FF,$00
.BYTE $35,$55,$FF
.BYTE $37,$7F,$F5
.BYTE $DD,$FF,$FC
.BYTE $D5,$FA,$BC
.BYTE $D7,$FF,$AD
.BYTE $F5,$EB,$EF
.BYTE $FF,$EB,$BD
.BYTE $D5,$FF,$AF
.BYTE $DD,$EA,$AD
.BYTE $DD,$FA,$FC
.BYTE $DD,$FF,$F0
.BYTE $37,$55,$FF
.BYTE $3D,$57,$D5
.BYTE $0F,$FF,$00
.BYTE $00,$00,$00
.BYTE $00,$00,$00
.BYTE $00,$00,$00
```

BASE_COLOR:□ MULTI_COLOR0:□ MULTI_COLOR1:■

URIDIUM/MANTA/02

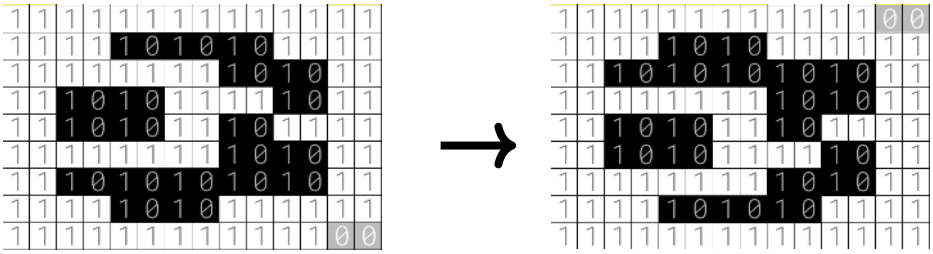
SPEC:MANTA_2
AZIMUTH:090°
OFFSET:$42
ELEVATION:000°
RAM:$5080
ROLL:015°



```
.BYTE $00,$00,$00
.BYTE $00,$00,$00
.BYTE $00,$00,$00
.BYTE $0F,$FF,$00
.BYTE $3D,$57,$FF
.BYTE $37,$55,$F5
.BYTE $DD,$FF,$F0
.BYTE $DD,$FA,$FC
.BYTE $DD,$EA,$AD
.BYTE $D5,$FF,$AF
.BYTE $FF,$EB,$BD
.BYTE $F5,$EB,$EF
.BYTE $D7,$FF,$AD
.BYTE $D5,$FA,$BC
.BYTE $DD,$FF,$FC
.BYTE $37,$7F,$FF
.BYTE $35,$55,$D5
.BYTE $0F,$FF,$00
.BYTE $00,$00,$00
.BYTE $00,$00,$00
.BYTE $00,$00,$00
```

BASE_COLOR:□ MULTI_COLOR0:□ MULTI_COLOR1:■

Closer inspection reveals a simple trick that behind this detailing: the yellow and black regions have simply had their order reversed and are mirror images of each other. For example if we take a look at the region representing the cockpit:

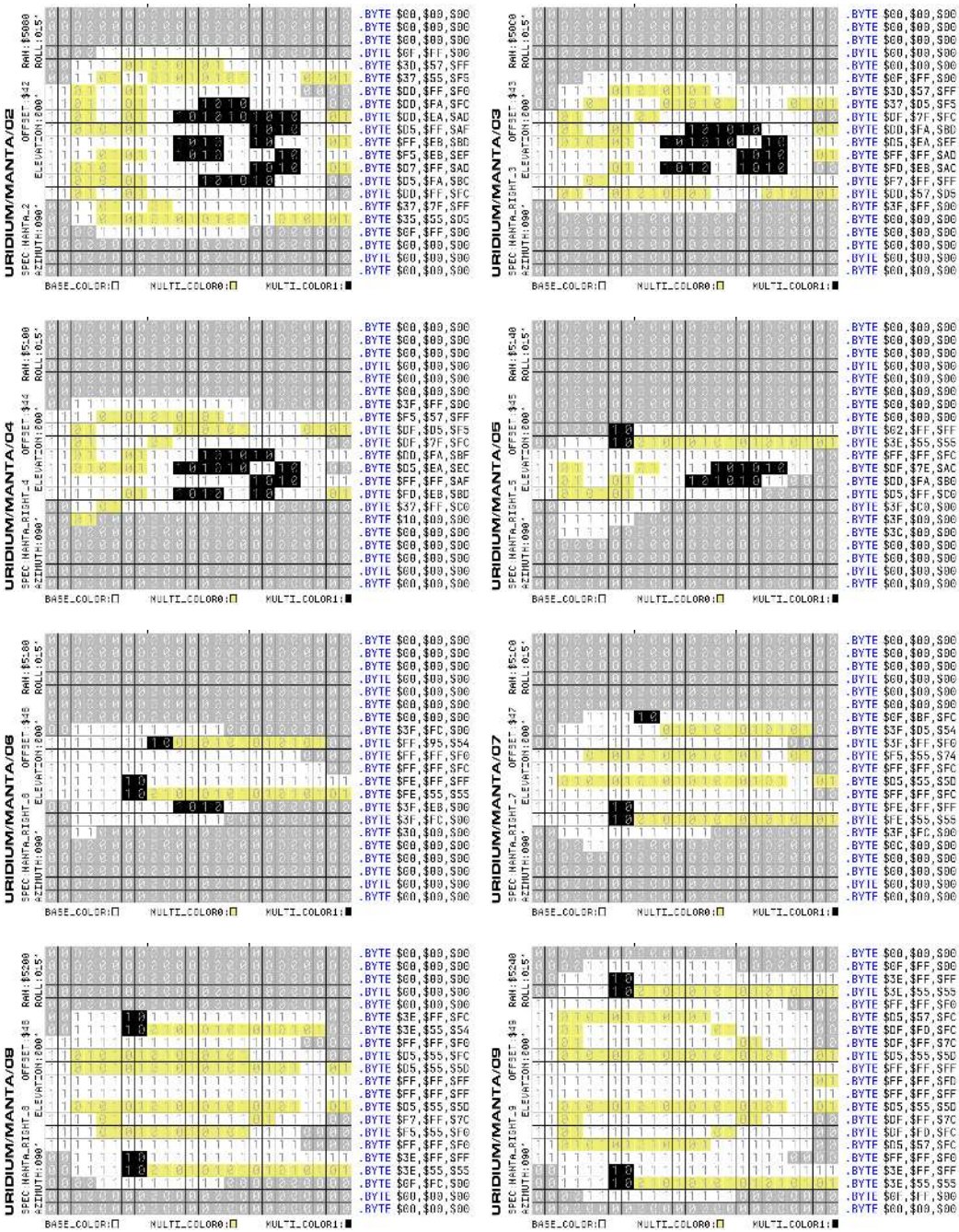


In its simplest terms the vertical order of the black lines detailing the cockpit have been reversed and this alone is enough to suggest the difference in attitude along the Y co-ordinate.

Examining this detail may also give you a sense of how the sprites themselves are defined. In the diagram above, for example, it may now be obvious that the black colour is defined by a bit pair of `10` and that the white colour is defined by a bit pair of `11`. In the diagrams opposite, we can further see that yellow is defined by the bit paid `01` and the transparent background of the sprite by `00`.

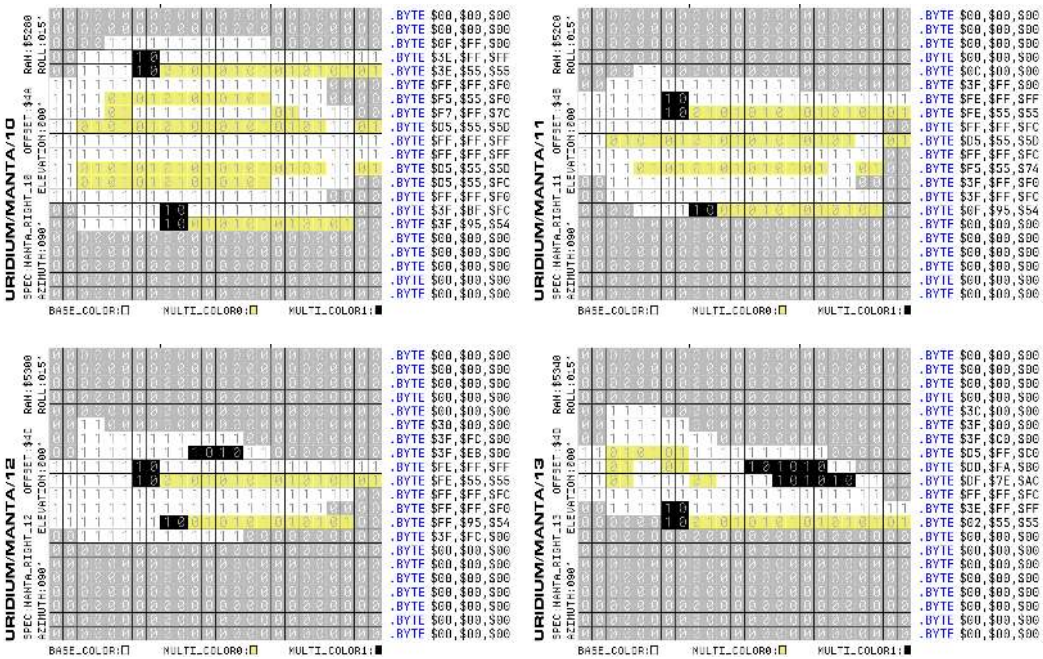
In the diagrams opposite and elsewhere we have given the three bytes that together make up the 12 bit pairs constituting each of the 21 lines in the sprite, a total of 63 bytes in total. This is the source data used by *Uridium's* 6502 assembler code to build each sprite using the straightforward process of mapping each bit pair to a corresponding pair of coloured pixels. An important feature of this sprite data is that it is stored contiguously for all sprites in the computer's RAM, or memory area, starting at address `$5000`. This is important because the order in which the sprite data is stored corresponds exactly to the order of each frame in the manta spin animation.

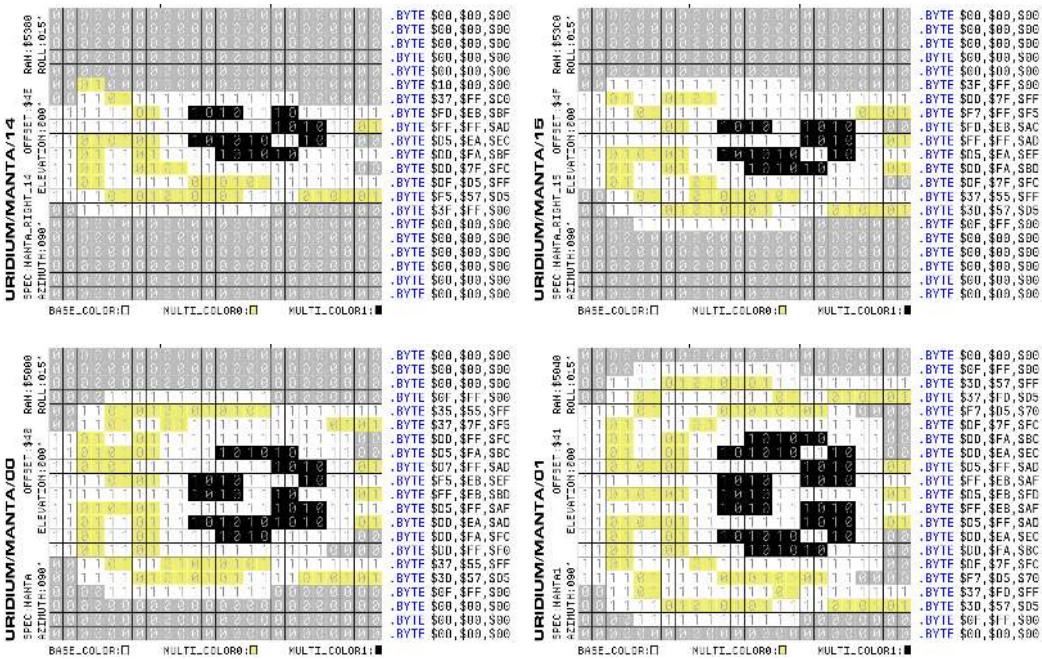
We can get a better sense of this by observing the RAM address of each sprite in the section of animation sequence between the ship faced towards and faced away. The first sprite in the sequence is at address `$5040`, the second at `$5080`, and so on. At each step, to get the next frame in the animation all that has to be done is to increment the position in RAM we are adressing by `$40`.


URIDIUM/MANTA/02
SPEC MANTA_2 OFFSET $42 RAH:$5080
AZIMUTH:090° ELEVATION:200° ROLL:0.5°
BASE_COLOR: MULTI_COLOR0: MULTI_COLOR1:
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYIL $0F,$FF,$90
.BYTE $3D,$57,$FF
.BYTE $37,$55,$F5
.BYTE $DD,$FF,$F0
.BYTE $DD,$FA,$FC
.BYTE $DD,$EA,$AD
.BYTE $D5,$FF,$AF
.BYTE $FF,$EB,$BD
.BYTE $F5,$EB,$EF
.BYTE $D7,$FF,$AD
.BYTE $D5,$FA,$BC
.BYIE $DD,$FF,$FC
.BYTE $37,$7F,$FF
.BYTE $35,$55,$D5
.BYTE $0F,$FF,$90
.BYTE $00,$09,$90
.BYIL $00,$09,$90
.BYTE $00,$09,$90
URIDIUM/MANTA/03
SPEC MANTA_RIGHT_3 OFFSET $43 RAH:$50C0
AZIMUTH:090° ELEVATION:200° ROLL:0.5°
BASE_COLOR: MULTI_COLOR0: MULTI_COLOR1:
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYIL $00,$09,$90
.BYTE $00,$09,$90
.BYTE $0F,$FF,$90
.BYTE $3D,$57,$FF
.BYTE $37,$D5,$F5
.BYIE $DF,$7F,$FC
.BYTE $DD,$FA,$BD
.BYTE $D5,$FA,$FF
.BYTE $FF,$FF,$AD
.BYTE $FD,$EB,$AC
.BYTE $F7,$FF,$FF
.BYIE $DD,$57,$D5
.BYTE $3F,$FF,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYIL $00,$09,$90
.BYTE $00,$09,$90
URIDIUM/MANTA/04
SPEC MANTA_RIGHT_4 OFFSET $44 RAH:$5100
AZIMUTH:090° ELEVATION:200° ROLL:0.5°
BASE_COLOR: MULTI_COLOR0: MULTI_COLOR1:
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYIL $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $3F,$FF,$90
.BYTE $F5,$57,$FF
.BYTE $DF,$D5,$F5
.BYTE $DF,$7F,$FC
.BYTE $DD,$FA,$BF
.BYTE $D5,$EA,$EC
.BYTE $FF,$FF,$AF
.BYTE $FD,$EB,$BD
.BYIE $37,$FF,$C0
.BYTE $10,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYIL $00,$09,$90
.BYTE $00,$09,$90
URIDIUM/MANTA/05
SPEC MANTA_RIGHT_5 OFFSET $45 RAH:$5140
AZIMUTH:090° ELEVATION:200° ROLL:0.5°
BASE_COLOR: MULTI_COLOR0: MULTI_COLOR1:
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYIL $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYIE $C2,$FF,$FF
.BYTE $3E,$55,$55
.BYTE $FF,$FF,$FC
.BYTE $DF,$7E,$AC
.BYTE $DD,$FA,$B0
.BYTE $D5,$FF,$C0
.BYTE $3F,$C0,$90
.BYTE $3F,$09,$90
.BYTE $3C,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYIL $00,$09,$90
.BYTE $00,$09,$90
URIDIUM/MANTA/06
SPEC MANTA_RIGHT_6 OFFSET $46 RAH:$5180
AZIMUTH:090° ELEVATION:200° ROLL:0.5°
BASE_COLOR: MULTI_COLOR0: MULTI_COLOR1:
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYIL $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $3F,$FC,$90
.BYIE $FF,$95,$54
.BYIE $FF,$FF,$F0
.BYTE $FF,$FF,$FC
.BYTE $FE,$FF,$FF
.BYTE $FE,$55,$55
.BYTE $3F,$EB,$90
.BYIE $3F,$FC,$90
.BYTE $30,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYIL $00,$09,$90
.BYTE $00,$09,$90
URIDIUM/MANTA/07
SPEC MANTA_RIGHT_7 OFFSET $47 RAH:$51C0
AZIMUTH:090° ELEVATION:200° ROLL:0.5°
BASE_COLOR: MULTI_COLOR0: MULTI_COLOR1:
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYIL $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $0F,$BF,$FC
.BYTE $3F,$D5,$54
.BYIE $3F,$FF,$F0
.BYIE $F5,$55,$74
.BYTE $FF,$FF,$FC
.BYTE $D5,$55,$50
.BYTE $FF,$FF,$FC
.BYTE $FE,$FF,$FF
.BYIE $FE,$55,$55
.BYTE $3F,$FC,$90
.BYTE $0C,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYIL $00,$09,$90
.BYTE $00,$09,$90
URIDIUM/MANTA/08
SPEC MANTA_RIGHT_8 OFFSET $48 RAH:$5200
AZIMUTH:090° ELEVATION:200° ROLL:0.5°
BASE_COLOR: MULTI_COLOR0: MULTI_COLOR1:
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYIL $00,$09,$90
.BYTE $00,$09,$90
.BYTE $3E,$FF,$FC
.BYTE $3E,$55,$54
.BYTE $FF,$FF,$F0
.BYIE $D5,$55,$FC
.BYIE $D5,$55,$50
.BYTE $FF,$FF,$FF
.BYTE $FF,$FF,$FF
.BYTE $D5,$55,$50
.BYTE $F7,$FF,$7C
.BYIL $F5,$55,$F0
.BYTE $FF,$FF,$F0
.BYTE $3E,$FF,$FF
.BYTE $3E,$55,$55
.BYTE $0F,$FC,$90
.BYIL $00,$09,$90
.BYTE $00,$09,$90
URIDIUM/MANTA/09
SPEC MANTA_RIGHT_9 OFFSET $49 RAH:$5240
AZIMUTH:090° ELEVATION:200° ROLL:0.5°
BASE_COLOR: MULTI_COLOR0: MULTI_COLOR1:
.BYTE $00,$09,$90
.BYTE $0F,$FF,$90
.BYTE $3E,$FF,$FF
.BYIL $3L,$55,$55
.BYTE $FF,$FF,$F0
.BYTE $D5,$57,$FC
.BYTE $DF,$FD,$FC
.BYTE $DF,$FF,$7C
.BYIL $D5,$55,$5D
.BYTE $FF,$FF,$FF
.BYTE $FF,$FF,$FD
.BYTE $FF,$FF,$FF
.BYTE $D5,$55,$5D
.BYTE $DF,$FF,$7C
.BYIL $DF,$FD,$FC
.BYTE $D5,$57,$FC
.BYTE $FF,$FF,$F0
.BYTE $3E,$FF,$FF
.BYTE $3E,$55,$55
.BYIL $0L,$II,$90
.BYTE $00,$09,$90

This approach does not hold for the second half of the animation sequence, or at least not all of it. As we can see below, the final two frames of the sequence reset back to address $5000 to stitch in the start of the animation and complete the loop.

```
SpinningShipAnimationLoop
        JSR CheckInputMaybeUpdateDecal
        LDA firePressed
        BEQ SpinningShipAnimationOver
        LDA someKindOfFrameRate
        BEQ SpinningShipAnimationOver
        JSR MaybeShowPauseScreen
        JSR StoreSpriteContentColorAndPosition
        INC currentSpriteValue
        LDA currentSpriteValue
        CMP #MANTA_16
        BCC Repeat
        LDA #MANTA_00
        STA currentSpriteValue
Repeat  JSR UpdateSpriteContentAndPosition
        JMP SpinningShipAnimationLoop
```




URIDIUM/MANTA/14
SPEC MANTA_RIGHT_14 OFFSET $42 RAH:$5300
AZIMUTH:090° ELEVATION:200° ROLL:0.5°
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $10,$09,$90
.BYTE $37,$FF,$C0
.BYTE $FD,$EB,$BF
.BYTE $FF,$FF,$AD
.BYTE $D5,$EA,$EC
.BYTE $DD,$FA,$BF
.BYTE $DD,$7F,$FC
.BYTE $DF,$D5,$FF
.BYTE $F5,$57,$D5
.BYTE $3F,$FF,$00
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
BASE_COLOR:
MULTI_COLOR0:
MULTI_COLOR1:
URIDIUM/MANTA/15
SPEC MANTA_RIGHT_15 OFFSET $4F RAH:$53C0
AZIMUTH:090° ELEVATION:200° ROLL:0.5°
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $3F,$FF,$90
.BYTE $DD,$7F,$FF
.BYTE $F7,$FF,$F5
.BYTE $FD,$EB,$AC
.BYTE $FF,$FF,$AD
.BYTE $D5,$EA,$FF
.BYTE $DD,$FA,$BD
.BYTE $DF,$7F,$FC
.BYTE $37,$55,$FF
.BYTE $3D,$57,$D5
.BYTE $0F,$FF,$00
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
BASE_COLOR:
MULTI_COLOR0:
MULTI_COLOR1:
URIDIUM/MANTA/00
SPEC MANTA OFFSET $42 RAH:$5000
AZIMUTH:090° ELEVATION:200° ROLL:0.5°
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $00,$09,$90
.BYTE $0F,$FF,$00
.BYTE $35,$55,$FF
.BYTE $37,$7F,$F5
.BYTE $DD,$FF,$FC
.BYTE $D5,$FA,$BC
.BYTE $D7,$FF,$AD
.BYTE $F5,$EB,$EF
.BYTE $FF,$EB,$BD
.BYTE $D5,$FF,$AF
.BYTE $DD,$EA,$AD
.BYTE $DD,$FA,$FC
.BYTE $DD,$FF,$F0
.BYTE $37,$55,$FF
.BYTE $3D,$57,$D5
.BYTE $0F,$FF,$00
.BYTE $00,$09,$90
.BYTE $00,$09,$90
BASE_COLOR:
MULTI_COLOR0:
MULTI_COLOR1:
URIDIUM/MANTA/01
SPEC MANTA1 OFFSET $41 RAH:$5040
AZIMUTH:090° ELEVATION:200° ROLL:0.5°
.BYTE $00,$09,$90
.BYTE $0F,$FF,$00
.BYTE $3D,$57,$FF
.BYTE $37,$FD,$D5
.BYTE $F7,$D5,$70
.BYTE $DF,$7F,$FC
.BYTE $DD,$FA,$BC
.BYTE $DD,$EA,$EC
.BYTE $D5,$FF,$AD
.BYTE $FF,$EB,$AF
.BYTE $D5,$EB,$FD
.BYTE $FF,$EB,$AF
.BYTE $D5,$FF,$AD
.BYTE $DD,$EA,$EC
.BYTE $DD,$FA,$BC
.BYTE $DF,$7F,$FC
.BYTE $F7,$D5,$70
.BYTE $37,$FD,$FF
.BYTE $3D,$57,$D5
.BYTE $0F,$FF,$00
.BYTE $00,$09,$90
BASE_COLOR:
MULTI_COLOR0:
MULTI_COLOR1:

a random walk through the source code of two video games from the dawn of the computer age